focusing on large-range search in the initial phases, and fine-tuning towards the end.

- **PROs**: it should be preferred when the domain of the cost function is discrete, the cost function is not differentiable or step-shaped, and when there are many local minima.
- **CONs**: it should be avoided when local minima are in narrow “valleys”, because it becomes unlikely the algorithm will find them. If the cost function is bowl-shaped or anyway with few minima, gradient descent should be preferred.
- **Speed**: the algorithm is also quite slow, because it randomly searches the domain, often going in the wrong direction. Compared to gradient descent, which takes the steepest to minima, simulated annealing will need many more iterations to get to the same point (or, more likely, to a different point with comparable cost).
- **Approximated Solutions**: simulated annealing can’t guarantee to return the optimal solution: if sub-optimal solutions are not admissible, then a different algorithm should be used.
- **Configuration**: simulated annealing takes several parameters that control how the system is evolved and if transitions uphill are allowed. Finding the right configuration requires time, and needs to be done for each problem (there isn’t a set of values that performs well on all problems). If you plan to use simulated annealing on a problem, be aware that this is the hardest part.
- **TSP**: traveling salesman problem is a difficult problem to solve, but it is ubiquitous in logistics, planning and